

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279254

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Chiplunkar; S et al.

Modular Pole Assembly for Overhead Line Systems

Abstract

A modular pole assembly for overhead line systems includes a top insulator housing defined by a first end and a second end. A bottom insulator housing is defined by a third end and a fourth end. The third end of the bottom insulator housing is removably coupled to the second end of the top insulator housing. A chamber is defined by at least one wall extending perpendicularly from at least one of the second end of the top insulator housing and the third end of the bottom insulator housing. The chamber is structured to accommodate at least one sensor.

Inventors: Chiplunkar; S (Nashik, IN), Patil; Chetan Dilip (Nashik, IN), Narayan Varade; Bhalchandra (Nashik, IN), Jakate; Vikas (Nashik, IN), Malik; Shamim (Nashik, IN), Trivedi; Pooja (Vadodara, IN)

Applicant: ABB Schweiz AG (Baden, CH)

Family ID: 84488161

Assignee: ABB Schweiz AG (Baden, CH)

Appl. No.: 19/210035

Filed: May 16, 2025

Related U.S. Application Data

parent WO continuation PCT/IB2022/061022 20221116 PENDING child US 19210035

Publication Classification

Int. Cl.: H01H33/662 (20060101); G01R31/12 (20200101); H02G7/05 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The instant application claims priority to International Patent Application No. PCT/IB2022/061022, filed Nov. 16, 2022, which is incorporated herein in its entirety by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to a modular pole assembly and, more particularly, to a configuration of the modular pole assembly that is adaptable to an overhead line system.

BACKGROUND OF THE INVENTION

[0003] Generally, a pole assembly consists of an interrupter along with current carrying components, which are either placed inside a housing or are molded together to form a pole assembly. The interrupter and the current carrying components are placed inside the housing and assembled with connecting parts. The vacuum interrupter may be operated by an actuator for selectively allowing or restricting the flow of current through the pole assembly.

[0004] Further, the pole assembly may be used in overhead line systems. The overhead line system may be circuit breakers, auto recloser or vacuum switch disconnectors. The overhead line system may include the pole assembly. The pole assembly may be of a live tank set up or of a dead tank set up. The pole assemblies may include insulating jackets that are molded to the outer surface in the live tank pole assembly set up. Further, the dead tank set up may directly accommodate the pole assembly in a dead potential casing.

[0005] Conventionally, pole assemblies were individually or separately manufactured for live tank configuration and dead tank configuration. The pole assemblies also include a sensor for measuring the voltage across the pole assembly. The sensor in conventional pole assemblies is molded or integrated with the pole assembly. Consequently, any damage to the sensor resulted in complete replacement of the pole assembly. Therefore, the overall operational and maintenance costs are increased, which is undesired.

BRIEF SUMMARY OF THE INVENTION

[0006] One or more shortcomings of conventional systems are overcome, and additional advantages are provided through an assembly as claimed in the present disclosure. Additional features and advantages are realized through the techniques of the present disclosure. Other embodiments and aspects of the disclosure are described in detail herein and are considered a part of the claimed disclosure.

[0007] In a non-limiting embodiment of the disclosure, a modular pole assembly for overhead line systems is disclosed. The modular pole assembly includes a top insulator housing defined by a first end and a second end. A bottom insulator housing is defined by a third end and a fourth end where the third end of the bottom insulator housing is removably coupled to the second end of the top insulator housing. A chamber is defined by at least one wall extending perpendicularly from at least one of the second end of the top insulator housing and the third end of the bottom insulator housing. Further, the chamber is structured to accommodate at least one sensor.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0008] FIG. 1 is a perspective view of a modular pole assembly in a live tank configuration in

accordance with an embodiment of the disclosure.

[0009] FIG. 2 is a sectional view of the modular pole assembly of FIG. 1.

[0010] FIG. 3 is an exploded view of the modular pole assembly, in accordance with an embodiment of the disclosure.

[0011] FIG. 4 is a magnified view of a portion of the modular pole assembly of FIG. 3.

[0012] FIG. 5 is a perspective view of a sensor of the modular pole assembly, in accordance with an embodiment of the disclosure.

[0013] FIG. 6 is a sectional view of the sensor of the modular pole assembly, in accordance with an embodiment of the disclosure.

[0014] FIG. 7 is a perspective view of a silicon jacket of the modular pole assembly, in accordance with an embodiment of the disclosure.

[0015] FIG. 8 is a sectional view of the silicon jacket of FIG. 7.

[0016] FIG. 9 is a perspective view of the modular pole assembly without the silicon jackets, in accordance with an embodiment of the disclosure.

[0017] FIG. 10 is a perspective view of the modular pole assembly in a dead tank configuration, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0018] The following paragraphs describe the present disclosure with reference to FIGS. 1 to 10. In the figures, the same element or elements which have similar functions are indicated by the same reference signs. For promoting an understanding of the principles of the disclosure, reference will now be made to specific embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended, such alterations and further modifications in the illustrated methods, and such further applications of the principles of the invention as illustrated therein being contemplated as would normally occur to one skilled in the art to which the disclosure pertains.

[0019] Reference is made to FIGS. 1-4, which illustrate a modular pole assembly **100** [herein after referred to as the pole assembly]. The pole assembly **100** may include a top insulator housing **102** and a bottom insulator housing **104**. The top insulator housing **102** may be defined by at least a wall which may further define a hollow space. The top insulator housing **102** includes a first end **116** and a second end **168**. The top insulator housing **102** may also be defined with an inner surface **102i** and an outer surface **1020**. The first end **116** may be a top end of the top insulator housing **102** and the second end **168** may be a bottom end of the top insulator housing **102**. The first end **116** of the top insulator housing **102** may be defined by an opening. The opening may be defined at the first end **116** along a substantially central section of the first end **116**. The opening may be configured to removably accommodate the first terminal **118**. The first terminal **118** may be configured in such a manner that a portion of the top insulator housing **102** may extend into a hollow space defined by the top insulator housing **102**. Further, the second end **168** of the top insulator housing **102** may be defined as an open end. A region of the second end **168** may be defined by a first threaded section **172**.

[0020] Particularly, the outer surface **1020** at the second end **168** of the top insulator housing **102** may be defined with the first threaded section **172**. In this preferable and non-limiting embodiment, the first threaded section **172** may be defined along the outer surface **1020** of the top insulator housing **102** such that, the first threaded section **172** may extend parallel to a first axis (A-A') of the pole assembly **100**. Further, the second end **168** may also include a bottom surface **146** that extends in a direction that is perpendicular to the first threaded section **172** and the first axis (A-A') of the pole assembly **100**. The bottom surface **146** may be configured above the first threaded section **172** and may also be defined along the outer surface **1020** of the top insulator housing **102**. The bottom surface **146** of the top insulator housing **102** may be defined with a semi-circular cutout [not shown in figures]. The semi-circular cutout may be configured to extend along the circumference or periphery of the bottom surface **146** and the top insulator housing **102**. Further, the hollow space of

the top insulator housing **102** may accommodate an interrupter **108** in an illustrated embodiment. The interrupter **108** may be housed in the hollow space such that, the interrupter **108** abuts the inner surface **102i** of the top insulator housing **102**. The interrupter **108** housed in the top insulator housing **102** may partially extend beyond the second end **168**.

[0021] One end of the first terminal **118** may be connected to a power source and may be configured to receive electricity and the opposite end of the first terminal **118** may be conductively connected to a top end of the interrupter **108**. The interrupter **108** in this non-limiting embodiment may be configured with a fixed conductor [not shown] and a movable conductor **108m**. The movable conductor **108m** may be selectively traversed to engage and disengage with the fixed conductor. The interrupter **108** may be configured to transmit the electricity from the first terminal **118** when the movable conductor **108m** is in contact with the fixed conductor. This condition may herein be termed as an engaged condition of the interrupter **108**. Further, the transfer of electricity is terminated when the movable conductor **108m** is traversed away from the fixed conductor and the same is termed as a disengaged condition of the interrupter **108**. In this non-limiting embodiment, the movable conductor **108m** is accommodated along a bottom region of the interrupter **108** and the movable conductor **108m** may extend into the bottom insulator housing **104**.

[0022] The bottom insulator housing **104** may also be defined by a wall which defines a hollow space. The bottom insulator housing **104** may be defined with a third end **166** and a fourth end **152**. The bottom insulator housing **104** may also be defined with an inner surface **104i** and an outer surface **1040**. The third end **166** may be a top end of the bottom insulator housing **104** and the fourth end **152** may be a bottom end of the bottom insulator housing **104**. The fourth end **152** of the bottom insulator housing **104** may be enclosed and the third end **166** of the bottom insulator housing **104** may be defined as an open end. The fourth end **152** of the bottom insulator housing **104** may include a second sealing ring **150**. The second sealing ring **150** may extend between the inner surfaces **104i** of the bottom insulator housing **104**. The second sealing ring **150** may prevent moisture and other dust particles from seeping into the hollow region of the bottom insulator housing **104**.

[0023] A region of the third end **166** may be defined by a second threaded section **174**. Particularly, the outer surface **1040** at the third end **166** of the bottom insulator housing **104** may be defined with the second threaded section **174**. In this preferable and non-limiting embodiment, the second threaded section **174** may be defined along the outer surface **1040** of the bottom insulator housing **104** such that, the second threaded section **174** extends parallel to the first axis (A-A') of the pole assembly **100**. Further, the third end **166** may also include a top surface **148** that extends in a direction that is perpendicular to the second threaded section **174** and the first axis (A-A') of the pole assembly **100**. The top surface **148** may be configured above the second threaded section **174** and may also be defined along the inner surface **104i** of the bottom insulator housing **104**. The top surface **148** of the bottom insulator housing **104** may be defined with a semi-circular cutout [not shown in figures]. The semi-circular cutout may be configured to extend along the circumference or periphery of the top surface **148** and the bottom insulator housing **104**.

[0024] The top insulator housing **102** and the bottom insulator housing **104** may be removably coupled to each other. The top insulator housing **102** and the bottom insulator housing **104** in this preferable and non-limiting embodiment may be coupled to each other through the first threaded section **172** and the second threaded section **174**. The first threaded section **172** and the second threaded section **174** may be defined with threads that engage with each other. The top insulator housing **102** may initially be positioned within the bottom insulator housing **104**. Further, one of the top insulator housing **102** and the bottom insulator housing **104** may be rotated with respect to the other. The first threaded section **172** engages with the second threaded section **174** and secures the top insulator housing **102** to the bottom insulator housing **104**. Further, the bottom surface **146** and the top surface **148** are oriented such that the semi-circular cut out of the bottom surface **146**

and the top surface **148**, together define a second groove **144**. The second groove **144** may extend between the second end **168** and the third end **166** of the top insulator housing **102** and the bottom insulator housing **104**, respectively. The second groove **144** may be configured to accommodate a first sealing ring **142**. The first sealing ring **142** may facilitate an airtight coupling between the top insulator housing **102** and the bottom insulator housing **104**. The above configuration at the second end **168** and the third end **166** enables the first sealing ring **142** to be accommodated in a manner which prevents moisture from seeping into the hollow regions of the top insulator housing **102** and the bottom insulator housing **104**. The above configuration at the second end **168** and the third end **166** with the first threaded section **172** and the second threaded section **174** respectively, may also enable an easier removal and coupling of the top insulator housing **102** and the bottom insulator housing **104**. In a preferable and non-limiting embodiment, the top insulator housing **102** and the bottom insulator housing **104** are made of material including but not limited to thermoclastic material. In a preferable and non-limiting embodiment, the interrupter **108** may be any device which is operable for selectively conducting electricity, including but not limited to vacuum interrupter. In an embodiment, the top insulator housing **102** and the bottom insulator housing **104** may be coupled to each other through means including but not limited to threads, snap fit arrangement etc.

[0025] The hollow space defined by the walls of the top insulator housing **102** and the bottom insulator housing **104** may accommodate a first conductor **132** and a second conductor **134**. The first conductor **132** may be a hollow elongated structure and the first conductor **132** may be conductively coupled to a bottom end of the interrupter **108**. Particularly, the first conductor **132** may be conductively coupled to the movable conductor **108m** of the interrupter **108**. The bottom end of the movable conductor **108m** of the interrupter **108** may be accommodated inside the hollow region of the first conductor **132**. The movable conductor **108m** may abut an inner surface **132a** of the first conductor **132** and the movable conductor **108m** may be fixedly coupled to the first conductor **132**. The first conductor **132** may be oriented to lie along the first axis (A-A'). Particularly, the first conductor **132** may be oriented such that the hollow section of the first conductor **132** extends through the first axis (A-A') of the pole assembly **100**. The first conductor **132** may be configured to move with the movable conductor of the interrupter **108**. The hollow space defined by the walls of the bottom insulator housing **104** may also accommodate an actuator **170**. One end of the actuator **170** may be accommodated on the fourth end **152** of the bottom insulator housing **104** and the opposite end of the actuator **170** may be conductively coupled to a bottom end of the interrupter **108**. As seen from FIG. 4, one end of the actuator **170** may extend into the hollow region of the first conductor **132** and the actuator **170** may be fixedly coupled to the movable conductor of the interrupter **108**. The actuator **170** may be selectively operated to impart a vertical movement to the first conductor **132** and the movable conductor of the interrupter **108**. Thus, the first conductor **132** is traversable along the first axis (A-A') of the pole assembly **100**. The actuator **170** in a non-limiting embodiment may be operated by any known means including but not limited to an electric motor, a hydraulic means etc. In an embodiment of the disclosure, the hollow space in the top insulator housing **102** and the bottom insulator housing **104** may accommodate components including but not limited to the interrupter **108** and the actuator **170**.

[0026] The pole assembly **100** may also include a second conductor **134**. The second conductor **134** may also be a hollow elongated structure that is defined in a shape including but not limited to that of a tubular structure. The second conductor **134** may be defined by an inner surface **134a** and an outer surface **134b**. The inner surface **134a** of the second conductor **134** may be defined with at least one first groove **138** [hereinafter referred to as the first groove]. The first groove **138** may extend throughout the circumference or periphery along inner surface **134a** of the second conductor **134**. The first groove **138** may be defined to extend along a plane that is perpendicular to the first axis (A-A'). In this preferable embodiment, at least two first grooves **138** may be defined along the inner surface **134a** of the second conductor **134**. Both the first grooves **138** may be defined along

planes that are parallel to each other. Further, the planes of both the first grooves **138** that are parallel to each other may also be oriented perpendicular to the first axis (A-A') of the pole assembly **200**.

[0027] The pole assembly **100** may also include at least one conductor plug **136** [hereinafter referred to as the conductor plug]. The conductor plug **136** may be positioned within the first groove **138**. In this preferable embodiment, the conductor plug **136** may be in the shape of a ring and the conductor plug **136** may be configured to abut the inner surface **132a** of the first conductor **132**. In this preferable and exemplary embodiment, the conductor plug **136** and the second conductor **134** may be of electrically conductive material. The diameter of the second conductor **134** measured from the second conductor **134** to the inner surface **134a** of the second conductor **134** may be equivalent to slightly greater than the diameter of the first conductor **132** that is measured from the center of the first conductor **132** to the inner surface of the first conductor **132**. The first conductor **132** and the second conductor **134** may be coupled to each other such that the inner surface **134a** of the second conductor **134** abuts the outer surface **132b** of the first conductor **132**. Further, the second conductor **134** may be fixedly connected to the bottom insulator housing **104** and the outer surface **134b** of the second conductor **134** may abut the inner surface **104i** of the bottom insulator housing **104**. As the actuator **170** is operated, the first conductor **132** may traverse vertically along the first axis (A-A') and the second conductor **134** may remain fixed. The outer surface **132b** of the first conductor **132** abuts the conductor plug **136** positioned in the first groove **138** of the second conductor **134**. The conductor plug **136** that is structured as the spring enables the conduction of high voltages and the above configuration also hinders/reduces the risk of electrical breakdown during conduction of electricity from the first conductor **132** to the second conductor **134**.

[0028] The pole assembly **100** may also include a chamber **106** that extends perpendicularly from at least one of the second end **168** of the top insulator housing **102** and the third end **166** of the bottom insulator housing **104**. The chamber **106** may be defined by walls that extend from at least one of the top insulator housing **102** and the bottom insulator housing **104**. The chamber **106** may also be defined with an outer surface **1060**. In this preferable and non-limiting embodiment, the chamber **106** may be configured to extend from the bottom insulator housing **104**. The chamber **106** may be configured to extend along a second axis (A-A') which is perpendicular to the first axis (A-A') of the pole assembly **100**. Further, the chamber **106** may be configured to extend from the region of the bottom insulator housing **104** that lies proximal or adjacent to the second conductor **134** in the bottom insulator housing **104**. The chamber **106** may further be defined by a proximal end **164** and a distal end **126**. The proximal end **164** of the chamber **106** may be the region that extends from the bottom insulator housing **104**. Further, a flange **124** may extend from the distal end **126** of the chamber **106**. The flange **124** may extend outwardly from the chamber **106** and the flange **124** may be defined with a diameter that is greater than the diameter of the walls defining the chamber **106**. The flange **124** may further be defined by an inner surface **124i** and an outer surface **1240**.

[0029] The pole assembly **100** further includes at least one sensor **128** [hereinafter referred to as the sensor]. The sensor **128** may be configured to measure the voltage in the pole assembly **100**. Reference is made from FIG. 4 to FIG. 6. The sensor **128** may include an extension **158** that extends from the surface of the sensor **128**. The extension **158** may be a plate-like structure that is housed or fixedly accommodated substantially on a central region of the sensor **128**. The extension **158** may be defined with a third groove **156**. The third groove **156** may be a cut out that extends along the circumference of the extension **158**. The sensor **128** may include a third conductor **130**. The third conductor **130** may be a hollow tubular shaped structure defined with a second cavity **130c**. The second cavity **130c** may be hole that extends through out the length of the sensor **128**. Further, a bushing **162** is housed within the chamber **106** and the bushing **162** is positioned adjacent to the proximal end **164** of the chamber **106**. The bushing **162** may also be defined with a

central hole that extends throughout the length of the bushing **162**. The shape of the bushing **162** may be defined such that an outer surface of the bushing **162** abuts an inner surface of the chamber **106**. Further, the hole extending throughout the center of the bushing **162** may define an inner surface which complements the shape of the sensor **128**. For instance, the hole in the bushing **162** may be defined with a semi conical shape which complements that semi conical shape of the sensor **128**. The sensor **128** may be accommodated within the bushing **162** such that the outer surface of the sensor **128** abuts the inner surface of the bushing **162**.

[0030] The sensor **128** is accommodated in the chamber **106** such that the second cavity **130c** of the third conductor **130** extends along the second axis (B-B) of the pole assembly. Further, the sensor **128** is accommodated in the bushing **162** such that one end of the third conductor **130** abuts the second conductor **134** in the bottom insulator housing **104**. The third conductor **130** of the sensor **128** may be coupled to the second conductor **134** such that the electricity from the second conductor **134** is conducted to the third conductor **130**. Further, region of the second conductor **134** that lies adjacent to the third conductor **130** and along the second axis (B-B) may be defined with a threaded hole. The threaded hole may be defined in the second conductor **134** such that the threaded hole is an extension of the second cavity **130c** in the third conductor **130**. Further, a fastener **140** may be inserted into the second cavity **130c** and the same may be engaged by threads with the threaded hole in the second conductor **134**. Thus, the sensor **128** may be secured with the second conductor **134** through the fastener **140**. The above configuration of coupling the sensor **128** with the second conductor **134** through the fastener **140** and the threaded hole must not be considered as a limitation and other configurations including but not limited to snap fit arrangements may be used. Further, the extension **158** of the sensor **128** may be positioned to abut or lie adjacent to the flange **124** of the chamber **106**. The extension **158** of the sensor **128** may be configured to abut the inner surface **124i** of the flange **124** such that the third groove **156** lies adjacent to the inner surface **124i** of the flange **124**. The third groove **156** may further accommodate a third sealing ring **154** and the third sealing ring **154** is structured to abut an inner surface **160** of the flange **124**. The above configuration at the third groove **156** in the extension **158** enables the third sealing ring **154** to be accommodated in a manner which prevents moisture from seeping into the chamber **106**.

[0031] Reference is made to FIG. 7 and FIG. 8 which illustrate a silicon jacket **114** for the chamber **106**. The silicon jacket **114** may be configured or provided on the outer surface **1060** of the chamber **106**. The silicon jacket **114** may be provided to close the first cavity **106a** of the chamber **106**. The silicon jacket **114** may be defined with a central cavity for accommodating a second terminal **122**. The second terminal **122** may partially protrude out from the silicon jacket **114**. Further, this silicon jacket **114** may be configured to enclose the chamber **106** and the sensor **128** accommodated in the chamber **106**. At least a portion of the silicon jacket **114** abuts the outer surface **1240** of the flange **124** at the distal end **126** of the chamber **106**. The silicon jacket **114** may be removably coupled to the flange **124** such that the region of the silicon jacket **114** that sits on the outer surface **1240** of the flange **124** is slidable over the outer surface **1240** of the flange **124**. Further, the second terminal **122** in the silicon jacket **114** may be configured to be accommodated within the second cavity **130c** of the third conductor **130** when the silicon jacket **114** is assembled to enclose the chamber **106**. The second terminal **122** may be configured to receive electricity from the third conductor **130**. The above configuration of the sensor **128** with the removable silicon jacket **114** enables an easy replacement of the sensor **128** when damaged. The silicon jacket **114** may be initially removed and the fastener **140** may be removed for disconnecting and replacing the sensor **128** from the second conductor **134**. Consequently, the cost of repairing the pole assembly **100** is significantly reduced as the sensor **128** may be individually replaced from the pole assembly **100**. The top insulator housing **102** and the bottom insulator housing **104** may also be configured with silicon jackets **110**, **112**. The silicon jackets **110**, **112** may be configured in a removable manner to the top insulator housing **102** and the bottom insulator housing **104**. The above

configuration of the pole assembly **100** with the silicon jackets **110**, **112** and **114** may be for one of live tank overhead line system including but not limited to a live tank circuit breaker, a live tank auto recloser, a live tank vacuum switch disconnecter.

[0032] Reference is further made to FIG. **9** and FIG. **10** that illustrate a dead tank. The pole assembly **100** without the silicon jacket **110**, **112**, **114** may be for one of dead tank overhead line system including but not limited to a dead tank circuit breaker, a dead tank auto recloser, a dead tank vacuum switch disconnecter. As seen from FIG. **10**, the pole assembly **100** may be accommodated in an insulated casing **178**. The configuration of the pole assembly **100** in the dead tank may be similar to the configuration of the pole assembly **100** in the live tank. Further, the chamber **106** may be configured to accommodate an adapter **176**. The adapter **176** may be a conductive element that is positioned to abut the second conductor **134** in the chamber **106**. The sensor **128** may further be positioned adjacent to the adapter **176** in the chamber **106**. Therefore, the sensor **128** and the silicon jacket **114** protrude out of the casing **178** and accessible externally. Consequently, the sensor **128** may be externally replaced without dismantling the casing **178** or accessing the pole assembly **100** in the casing **178**.

[0033] In an embodiment, an arcing distance between the first terminal **118** and the second terminal **122** may be varied. The silicon jacket **114** on the chamber **106** may be partially moved along the second axis (B-B) and over the flange **124** such that the distance between the first terminal **118** and the second terminal **122** is varied. Thus, the arcing distance between first terminal **118** and the second terminal **122** may be suitably varied.

[0034] In an embodiment, the modular pole assembly **100** with the silicon jacket **110**, **112**, **114** accommodated on the top insulator housing **102**, the bottom insulator housing **104** and the chamber **106**, forms the live tank. The silicon jacket **110**, **112**, **114** may be effortlessly removed from the top insulator housing **102**, the bottom insulator housing **104** and the chamber **106** of the modular pole assembly **100**. The modular pole assembly **100** may be subsequently adapted or positioned in the casing **178** as seen from the FIG. **10** which forms the dead tank. Thus, the modular pole assembly **100** is adaptable to form the live tank or the dead tank by selectively removing or accommodating the silicon jacket **110**, **112**, **114**. Consequently, manufacturing costs for live tank and dead tank reduces and modular pole assembly **100** is easily adaptable to form the live tank or the dead tank. In an embodiment, the sensor **128** may also be easily replaced by removing the silicon jacket **114** on the chamber **106**. The modular pole assembly **100** is configured such that the sensor **128** is easily accessible by removing the silicon jacket **114** on the chamber **106**. Thus, a faulty or damaged sensor **128** may be individually replaced without having to replace the complete modular pole assembly **100**.

[0035] In an embodiment of the disclosure, an interrupter is disposed in the top insulator housing and extends into the bottom insulator housing.

[0036] In an embodiment of the disclosure, at least one silicon jacket is coupled to an outer surface of each of the top insulator housing, the bottom insulator housing and the chamber.

[0037] In an embodiment of the disclosure, the chamber is defined with a flange at a distal end of the chamber, and the flange is structured to support a portion of the at least one sensor.

[0038] In an embodiment of the disclosure, at least a portion of the at least one silicon jacket abuts an outer surface of the flange at the distal end of the chamber.

[0039] In an embodiment of the disclosure, a first conductor is conductively coupled to the interrupter.

[0040] In an embodiment of the disclosure, a second conductor conductively coupled to the first conductor wherein, an inner surface i of the second conductor is defined with at least one first groove.

[0041] In an embodiment of the disclosure, a conductor plug is accommodated in the at least one first groove, and the conductor plug is structured to an outer surface b of the first conductor.

[0042] In an embodiment of the disclosure, the at least one sensor is removably coupled by a

fastener, extending through a provision along a length of the at least one sensor and coupled with the second conductor.

[0043] In an embodiment of the disclosure, a first sealing ring accommodated in a second groove defined on a bottom surface of the top insulator housing and a top surface of the bottom insulator housing.

[0044] In an embodiment of the disclosure, a second sealing ring disposed to the fourth end of the bottom insulator housing.

[0045] In an embodiment of the disclosure, a third sealing ring disposed in a third groove defined on an extension of the at least one sensor wherein, the third sealing ring is structured to abut an inner surface of the flange.

[0046] In an embodiment of the disclosure, the modular pole assembly is of a thermoplastic material.

[0047] In an embodiment of the disclosure, a bushing is positioned at a proximal end in the chamber and accommodating the at least one sensor.

[0048] In an embodiment of the disclosure, the at least one silicon jacket is configured to close the first cavity as defined by the third chamber.

[0049] In an embodiment of the disclosure, the modular pole assembly with at least one silicon jacket forms a live tank and the modular pole assembly housed in a casing without the at least one silicon jacket defines a dead tank.

[0050] All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0051] The use of the terms “a” and “an” and “the” and “at least one” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The use of the term “at least one” followed by a list of one or more items (for example, “at least one of A and B”) is to be construed to mean one item selected from the listed items (A or B) or any combination of two or more of the listed items (A and B), unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

[0052] Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein.

Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

TABLE-US-00001 Referral numerals: Detailed description Referral numeral Modular pole assembly 100 Top insulator housing 102 Bottom insulator housing 104 Chamber 106 First cavity

106a Outer surface 106b Interrupter 108 Movable conductor 108m Silicon jacket 110 Silicon jacket 112 Silicon jacket 114 First end 116 First terminal 118 Outer surface of the second terminal 120 Second terminal 122 Flange 124 Outer surface of the flange 124a Inner surface of the flange 124b Distal end 126 Sensor 128 Third conductor 130 Second cavity 130c First conductor 132 Inner surface of the first conductor 132a Outer surface of the first conductor 132b Second conductor 134 Inner surface of the second conductor 134a Inner surface of the second conductor 134b Conductor plug 136 First groove 138 Fastener 140 First sealing ring 142 Second groove 144 Bottom surface of the top insulator housing 146 Top surface of the bottom insulator housing 148 Second sealing ring 150 Fourth end 152 Third sealing ring 154 Third groove 156 Extension 158 Bushing 162 Proximal end 164 Third end 166 Second end 168 Actuator 170 First threaded section 172 Second threaded section 174 Adapter 176 Casing 178 Pole assembly 100

Claims

1. A modular pole assembly for overhead line systems, comprising: a top insulator housing having a first end and a second end; a bottom insulator housing having a third end and a fourth end, wherein the third end is removably coupled to the second end; and at least one wall extending perpendicularly from at least one of the second end of the top insulator housing and the third end of the bottom insulator housing, the at least one wall defining a chamber structured to accommodate at least one sensor.
2. The modular pole assembly of claim 1, further comprising an interrupter disposed in the top insulator housing and extending into the bottom insulator housing.
3. The modular pole assembly of claim 1, further comprising at least one silicon jacket coupled to an outer surface of each of the top insulator housing, the bottom insulator housing, and the chamber.
4. The modular pole assembly of claim 3, wherein the chamber is further defined by a flange disposed at a distal end of the chamber, the flange being structured to support a portion of the at least one sensor.
5. The modular pole assembly of claim 4, wherein at least a portion of the at least one silicon jacket abuts an outer surface of the flange at the distal end of the chamber.
6. The modular pole assembly of claim 2, further comprising a first conductor conductively coupled to the interrupter.
7. The modular pole assembly of claim 6, further comprising a second conductor conductively coupled to the first conductor, wherein an inner surface of the second conductor is defined with at least one first groove.
8. The modular pole assembly of claim 7, further comprising a conductor plug accommodated in the at least one first groove, wherein the conductor plug is structured to an outer surface of the first conductor.
9. The modular pole assembly of claim 7, wherein the at least one sensor is removably coupled by a fastener extending through a provision along a length of the at least one sensor and coupled with the second conductor.
10. The modular pole assembly of claim 1, further comprising a first sealing ring accommodated in a second groove defined on a bottom surface of the top insulator housing and a top surface of the bottom insulator housing.
11. The modular pole assembly of claim 1, further comprising a second sealing ring disposed on the fourth end of the bottom insulator housing.
12. The modular pole assembly of claim 4, further comprising a third sealing ring disposed in a third groove defined on an extension of the at least one sensor, wherein the third sealing ring is structured to abut an inner surface of the flange.
13. The modular pole assembly of claim 1, wherein at least a portion of the top and/or bottom

insulator housing is made of a thermoplastic material.

14. The modular pole assembly of claim 1, further comprising a bushing positioned at a proximal end in the chamber and accommodating the at least one sensor.

15. The modular pole assembly of claim 1, wherein the at least one silicon jacket is configured to close the cavity.

16. The modular pole assembly of claim 1, wherein the at least one silicon jacket forms a live tank.

17. The modular pole assembly of claim 1, further comprising a casing that defines a dead tank.
